

# Duty Monitor "Monitoring System using NFC"

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## ABSTRACT

Duty Monitor is a project aiming to develop an NFC tracker framework to screen the area of representatives in the working environment. Every worker will be given a cell phone with an implanted NFC tag, and certain areas will be set apart with an NFC peruser. This system will help organizations increase their working environment performance and allow administration to quickly locate staff when needed. The project includes designing, implementing, and deploying devices such as an NFC peruser, NFC labels, GUI, and an information base.

**KEYWORDS** : *IoT, GUI, Tracking system, NFC.*

## INTRODUCTION

Location and staff tracking systems using NFC (Near Field Communication) tags have gained significant attention due to the growing need for efficient management, enhanced security, and streamlined operations within organizations. This technology leverages the capabilities of tags to enable the real-time tracking and monitoring of employees' movements, attendance, and access control.

Personnel tracking systems using NFC bolster workplace security by ensuring that only authorized personnel can access certain areas of the organization. Unauthorized access is detected and can trigger alerts, enhancing overall safety. Traditional attendance systems often involve manual data entry and paperwork, leading to errors and inefficiencies. These tags enable real-time monitoring of officers, making it easier to locate staff during emergencies or in large facilities. This is invaluable for evacuations or responding to urgent situations. This can help enhance productivity by identifying bottlenecks in processes and promoting

accountability. Admin can assess where staff spend their time and allocate resources more effectively.

### Objectives

The objectives of this project are to solve the problems of monitoring police officers' duties and their presence at the workplace. As it is time-consuming and rigorous to get the exact location of staff. In an advancing nation like India, various most recent innovations have been grown like NFC, remote, Bluetooth, robots, etc. In this manner, these advancements can be utilized to guarantee the representative time in/out and to follow the area of the staff at explicit areas at work during work hours. This framework additionally facilitates the organization to screen their work force promptly from the record in the data set other than building a powerful framework that consequently records and updates the area of the faculty continuously. It will permit its boss to see changes when they happen, instead of trusting that updates will be noticeable at a future time.

Albeit Close to handle correspondence (NFC) is one of

the programmed distinguishing proof advances more stylish these days. There is a wide innovative work in this space attempting to exploit this innovation, and before long numerous new applications and examination regions will keep on showing up. This unexpected interest in NFC likewise achieves a few worries, for the most part about the security and protection of the people who work with or use labels in their regular day to day existence.

### Problem Definition

Sometimes the cops or employees do not follow their duty and it affects the law and order of society. If an officer moves away from their assigned post without proper authorization, then it is harmful for security. Manually it is not possible, so we come up with a solution that is Tracking police officers deployed to bandobast duty (security duty) remotely using NFC (Near Field Communication) technology to ensure they stay at their assigned posts. It will enhance the productivity and efficiency of the officers and also save the time of administration.

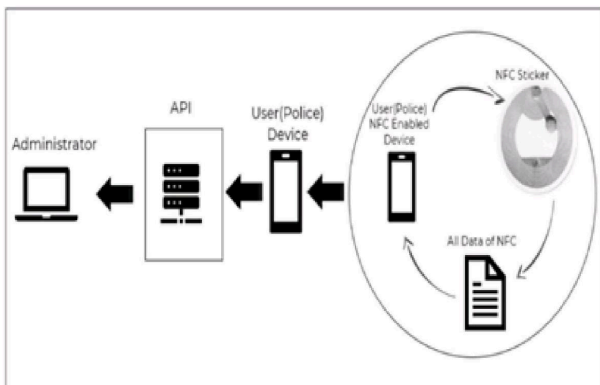


Fig. 1. NFC Architecture

### Scope of the Project

The project's scope is broken down into four primary components: users, functionality, operability of the system, and users of development tools. Employees To help the administrator keep track of the registered personnel, users are able to register and update their information in the system. Because they can add, edit, delete, view, and search people in the system in addition to looking up personnel locations, administrators are the most essential users in this system.

### Proposed Solution

NFC Check-In/Check-Out System:

- Equip each police officer with an NFC-enabled badge or card.
- Install NFC readers at strategic locations at assigned posts.
- Officers must tap their NFC badge on the reader when they arrive at their post (check-in) and when they leave (check-out).

Attendance Recording:

- Integrate the NFC data with a centralized attendance recording system.
- This system should log the time and location of each check-in and check-out, ensuring real-time tracking.

Geotagged Image capture:

- Require personnel to take geotagged images with their smartphone's assigned posts when they check in at their assigned posts.
- Implemented a system that automatically uploads these images to a secure server.
- Admin can view these images to verify the personnel's presence at the assigned location.

## REVIEW OF LITERATURE

### Survey of Existing System

In one review, the coordination of LBSs with NFC innovation is performed to follow clients' way of behaving and to further develop client encounters. Urban communities like Frankfurt and Oulu are instances of labeled conditions empowering cell phones to recover area explicit information from accessible labels. The labels might be appended to leave meters, transport stops, streetlights, or a few different areas; and may make a foundation for clients to look for data about the nearby administrations. Subsequent to tending to such models, the creators offer the making of an area based versatile Wiki. In addition, constant area information might upgrade person to person communication, where the proposed LocaTag framework utilizes texting devices, area information, and NFC innovation.

In another review, an NFC registration framework is introduced, which benefits from Suggested Area Based Administrations (ILBS), cell phones, and NFC labels. To empower a particular activity, the client needs to genuinely be at a particular area, permitting more close collaboration. Benefits of the suggested NFC registration administration concerning manual registrations (i.e., Facebook, Foursquare, etc) are introduced in a similar report.

#### Limitations on existing system

- The current system is unable to capture live photos on duty.
- The system can not work without an internet connection or saving data.
- Existing systems are costly and hard to maintain.

## ANALYSIS MODELLING

### Flowchart

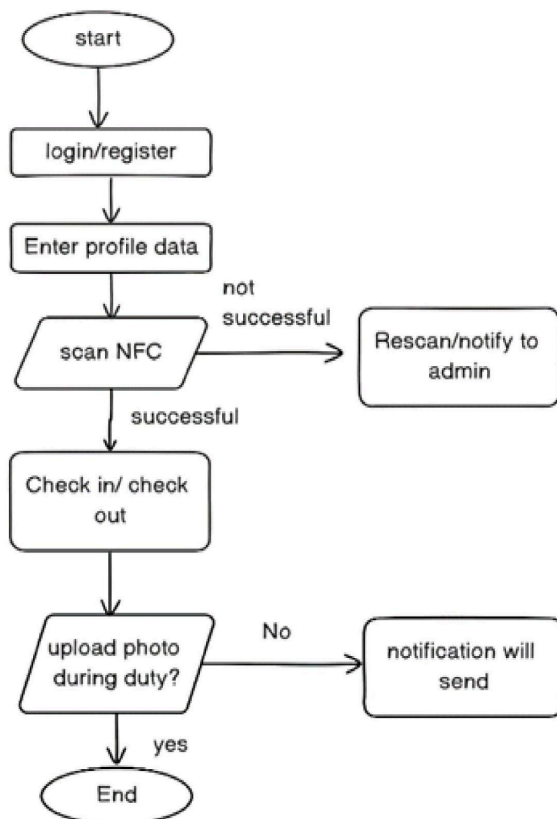


Fig. 2. Flow Diagram

### ER Diagram

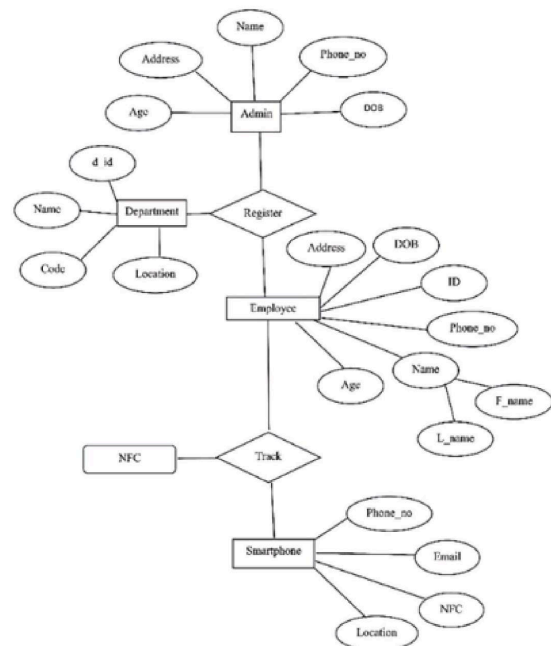


Fig. 3. Er Diagram

## CONCLUSION

In conclusion, utilizing NFC technology to implement a police officer tracking system can offer numerous benefits to administration, including enhanced security, increased efficiency, and simplified access control. At last, the outcome of an Obligation Screen framework relies upon cautious preparation, clear correspondence, and a pledge to adjusting the benefits of further developed effectiveness and security with the insurance of official's protection and information. When executed mindfully and dependably, NFC-Based on the Obligation Screen can be an important resource for organization and organizations.

## REFERENCES

1. KWANG. W. NAM 1, (Member, IEEE), AND KWANG. S YANG, (Member, IEEE) "reairoi: Discovering Real Regions of Interest From Geotagged Photos", date of current version 12 August 2022.
2. Emanuel Raso 1, Giulio Maria Bianco 2, Lorenz Braciale 1, Gaetan Maroco 2, Cecilia occhiuzzi 2, and Pierpalo Loreti 1, "Privacy-Aware Architectures for NFC and RFID Sensors in Healthcare Applications",

3. "An Introduction to Near Field Communication (NFC)", Published in Chip-Monks.
4. Selin Olinik, Hong Seok Lee and Firat Gilder, "The future of near-field communication-based wireless sensing"
5. MediumArticle: <https://medium.com/chip-monks/an-introduction-to-near-fieldcommunication-nfc-ad445a87078b>
6. Regarding NFC: <https://www.mdpi.com/1424-8220/22/24/9692>
7. NFC, Wikipedia, Available: [http://en.wikipedia.org/wiki/Near\\_field\\_communication](http://en.wikipedia.org/wiki/Near_field_communication)
8. NFC Security Available: <http://www.radio-electronics.com/info/wireless/nfc-near-field-communications-security.php>
9. Awais Mulla, Uzair Ansari, Shadab Khan, Employee Tracking System Using NFC, e-ISSN: 2395-0056, Volume: 05 Issue: 02 Feb-2018
10. Anusha Rahul, Gokul Krishnan G, Unni Krishnan H and Sethuraman Rao, (NFC) TECHNOLOGY. M. Pulipati and S. Phani, "Comparison of various short range wireless communication technologies with nfc," Interface J. Sci. Res, vol. 2, pp. 87–91, 2013.
12. Roland, Basics. Charm, Switzerland: Springer, 2015, pp. 13–31.
13. K. Shabana, N. Fidaa, F. Khan, S. Jan, and M. U. Rehman, "Security issues and attacks in wireless sensor networks," Int. J. Adv. Res. Computer Science Electron. Enginer
14. W. Yon and N. Vaidya, "NFC reader collision problem: Performance analysis and medium access," Wireless Commun. Mobile Comput., vol. 12, no. 5, pp. 420–430, Apr. 2012.
15. C. Mears. (2016). Why Organizations Need Adaptive Multi-Factor Authentication (MFA). Accessed: Nov. 20, 2018. [Online]. Available: <https://blog.centrify.com/adaptive-multi-factor-authentication-mfad>
16. MIT University. (2017). Kerberos: The Network Authentication Protocol. Accessed: Nov. 10, 2018. [Online]. Available: <https://web.mit.edu/krbros/>
17. SJB Research. NFCW. Accessed: Oct. 4, 2019
18. L. Mainetti, L. Patino, and R. Vergalo, "IDA-Pay: A secure and efficient micro-payment system based on peer-to-peer NFC technology for Android mobile devices," J. Comun. Software Syst., vol. 8, no. 4, pp. 117–125, 2012.
19. D. M. Montero, J. J. P. C. Rodrigs, and J. Lloret, "A secure NFC application for credit transfer among mobile phones," in Proc. Int. Conf. Computer., Inf. Telecommunication. System. (CITS), May 2012, pp. 1–5.